

Live Coding Exercise: CSF Biomarker Exploration

Scenario

BridgeBio is developing an antisense oligonucleotide (ASO) therapy targeting a CNS-expressed gene (Gene X) for a rare neurodevelopmental disease. The ASO is designed to reduce expression of Gene X in the central nervous system.

In this exercise, you are provided with **synthetic CSF omics data** from a small, Phase-1–style study:

- 20 subjects (≈12 ASO, 8 placebo)
- CSF collected at **baseline** and **week 4**
- For each CSF sample:
 - Study metadata (treatment, timepoint, batch, site, dose, etc.)
 - **CSF proteomics** (300 proteins, including the protein product of Gene X)
 - **CSF transcriptomics** (500 RNAs, including Gene X)

All data are pre-processed and stored in the `data/` directory.

Goal

Explore whether there are **molecular changes in CSF** that could serve as **biomarkers of target engagement** for the ASO.

This is intentionally a *larger* problem than can be fully completed during the interview. We are interested in **your reasoning, prioritization, and communication**, not a polished solution.

Helpful considerations:

- How you orient yourself in the data
 - How you form and refine hypotheses about ASO-related CSF shifts
 - How you choose comparisons or summaries to evaluate target-engagement signals
 - How you communicate what you see, what you trust, and what you'd do next
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Notes

- You are welcome to use any online/public resources you have available to you throughout this exercise.
- You are free to plot using **any library you prefer** (matplotlib and seaborn are already available).
- You may find it useful to examine **baseline** → **week 4** changes and **ASO vs placebo** differences, but your analytical approach is flexible.
- Target markers:
 - **GENE001** (RNA target)
 - **PROT001** (protein target)
- It is expected that you will only partway through the analysis.
The goal is to understand **your approach**, not your throughput.

A strong end state (which you will unlikely reach today) would be:

- A short list of **candidate target-engagement biomarkers**,
- With a brief justification (e.g., effect direction, treatment/timepoint pattern, link to mechanism).